

Research Directions on Kolmogorov–Arnold Networks: A Comprehensive Review

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Abstract

Utilizing the Kolmogorov–Arnold representation theorem, KANs have emerged as a mathematically rigorous and easily interpretable alternative to traditional neural networks. These networks decompose high-dimensional functions into sums of univariate continuous functions using adaptive activation functions. Compared to MLPs, KANs exhibit superior or comparable performance in accuracy, parameter efficiency, and interpretability. Applications highlight the advantages of KANs in solving complex partial differential equations with enhanced convergence and uncertainty quantification, modeling dynamic systems in a meaningful manner, and making reliable forecasts in the areas of power systems, environmental monitoring, and demand prediction. Based on current research on KANs, they demonstrate a promising frontier in interpretable deep learning, with increasing influence across numerous interdisciplinary scientific and engineering fields.

Keywords: Kolmogorov–Arnold networks; function approximation; neural network interpretability; hybrid deep architectures; scientific computing

1. Theoretical Foundations of Kolmogorov–Arnold Networks

Symmetry, as an intrinsic property of both natural and mathematical systems, enhance model generalizability. It also serves as a crucial bridge linking data-driven approaches to mechanistic modeling. This characteristic is also evident in the recently emerged KAN field. A KAN is theoretically based on the Kolmogorov–Arnold representation theorem, which states that any continuous multivariate function with a bounded domain can be reduced to a finite composition of univariate continuous functions and the binary operation of addition for any multivariate continuous function [1]. The theorem provides a constructive method of function approximation, suggesting that high-dimensional mappings can be broken down into simple univariate components, which ultimately influences KAN architecture by guiding the decomposition of complex functions into univariate function compositions. Making the neural network design have a clear mathematical grounding and fundamentally improving model structure interpretability.

This theorem led to KANs being developed as an alternative to MLPs. Hecht-Nielsen initially proposed shallow Kolmogorov networks based on the theorem, but their practical ineffectiveness arose from poor univariate function properties [2]. Later, Liu et al. deepened this framework to develop KANs, which integrate learnable activation functions to enhance performance, effectively combining the structural insights of Kolmogorov networks with the flexibility of MLPs [3]. Mathematically, KANs approximate functions as



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cascaded compositions of layers (Φ), where each layer Φ is a sum of univariate functions φ , often parameterized by basis functions (e.g., B-splines, polynomials) with learnable coefficients [4].

The structure of KANs differs fundamentally from the structure of MLPs. Based on linear transformations, traditional MLPs apply fixed nonlinear activations (e.g., Tanh, ReLU) at nodes, according to the following form [5]:

$$MLP(x) = (W_L \circ \sigma_{L-1} \circ \dots \circ W_1)(x) \quad (1)$$

Here, W denotes the weights between network layers, and σ represents the activation function. Contrary to KANs, the edges (weights) of KANs contain learnable univariate activation functions, while the nodes only perform summation operations. As a result, the computational graph has been shifted to

$$KAN(x) = (\Phi_L \circ \dots \circ \Phi_1)(x) \quad (2)$$

where Φ denotes matrices of univariate functions (e.g., B-splines, orthogonal polynomials) replacing scalar weights [6,7]. This edge-based activation design eliminates the need for separate linear weight matrices, integrating transformation and nonlinearity into a unified, interpretable layer structure [8]. Figure 1 shows the comparison between them.

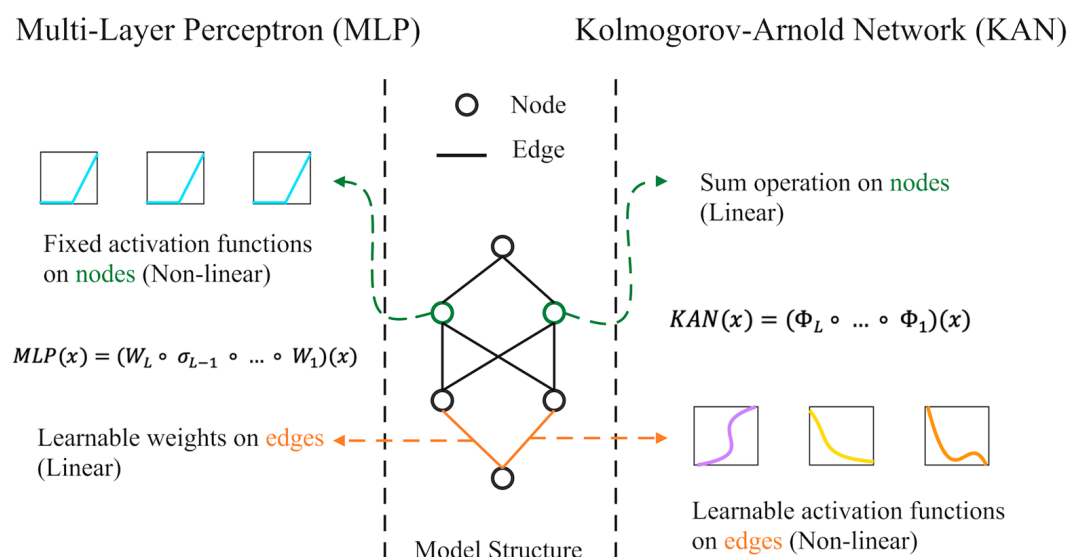


Figure 1. MLPs vs. KANs.

In terms of their approximation mechanism, MLPs differ functionally from KANs. A KAN, in contrast to MLPs, implements the Kolmogorov–Arnold decomposition explicitly by summing compositions of parameterized univariate functions, rather than using nested layers of linear combinations followed by fixed nonlinearities [9]. Accordingly, KANs benefit from guaranteed constructive representation under the theorem, which complements MLPs’ universal approximation capability in a theoretically grounded manner [10,11]. KANs’ activation functions are highly flexible; early implementations used B-splines (e.g., vanilla PIKAN), while variants adopt orthogonal polynomials (Chebyshev, Jacobi), radial basis functions, or fractional Jacobi polynomials to improve efficiency, stability, or adaptability [12].

Training strategy and parameter complexity further distinguish KANs from MLPs. During the training phase, MLPs adjust the linear weights on the edges while keeping the nonlinear activation functions on the nodes fixed; while KANs adapt the nonlinear

activation functions on the edges while keeping the linear operations on the nodes fixed. Due to the size of the grid and the polynomial order of the spline grids, KANs may be more complex. However, modified variants (e.g., Chebyshev-KAN) often reduce redundant parameters, enabling compact architectures with fewer nodes than MLPs [13]. By replacing linear layers with KAN-based function matrices, this structural and functional divergence distinguishes KANs from MLPs as an interpretable, theoretically grounded alternative. This paper makes the following contributions:

1. According to the latest research using KANs, 80 papers were selected for analysis and review based on journal impact factors, citation counts, publication years, and journal rankings.
2. This paper provides an overview of KAN applications across multiple domains. Analyzing the characteristics of these models enables readers to better apply and understand these models within their respective fields.
3. This paper contains comparative research on KANs. It enables readers to gain a deeper understanding of the KAN model, facilitating its more judicious application.
4. This paper explores existing challenges, limitations, and future research directions, providing relevant insights for subsequent research and development.

2. Architecture and Variants of KAN Models

Classical KANs are rooted in the Kolmogorov–Arnold representation theorem, decomposing multivariate functions into sums of learnable univariate activations on network edges, distinct from fixed node activations in traditional MLPs [14,15]. B-splines serve as the foundational activation of offering smooth, piecewise polynomial approximations. Their mathematical formulation typically combines a residual-like term with B-spline basis functions:

$$\varphi(x) = \omega(b(x) + spline(x)) \quad (3)$$

$$b(x) = silu(x) = x / (1 + e^{-x}) \quad (4)$$

$$spline(x) = \sum_i c_i B_i(x) \quad (5)$$

With ω scaling magnitude, $b(x)$ as a basis function (e.g., SiLU), c_i as trainable coefficients, and $B_i(x)$ as B-spline basis functions. This activation function is well-suited to image recognition [16] and time series data prediction [17].

An extension of the KAN architecture that integrates sinusoidal embeddings with MLP layers is the FCN-KAN. The function is defined as

$$\varnothing(x) = Linear(SiLU(Linear(concat(x, \sin(\omega x), \cos(\omega x))))) \quad (6)$$

In this equation, ω is a learnable coefficient representing the frequency of sinusoidal functions and x is the input value.

KAN layers are structured as matrices of such spline functions, enabling compositional multivariate mapping [18,19]. Computationally, B-splines support grid adaptation to enhance pattern capture [20], but incur higher complexity due to iterative basis calculations, limiting scalability in deep networks [21,22]. Practical implementations optimize spline order and grid points to balance performance and cost [23,24]. Furthermore, KANs reduce computational complexity by limiting node operations to simple additions and streamlining the computation process [25].

RBF-KAN is a quicker version of Spline-KAN that calculates the higher-order B-spline basis with Gaussian RBFs described as

$$\varnothing(r) = \exp\left(-\left(\frac{r}{h}\right)^2\right) \quad (7)$$

In this equation, r is the radial distance between input x and grid center points c_i , and it can be calculated as $r = \|x - c_i\|$. h denotes the width or extent of the Gaussian function specified by the grid range.

Polynomial basis functions, including orthogonal variants, offer alternatives to splines. Traditional polynomial KANs suffer from numerical instability (e.g., Runge's phenomenon), but orthogonal polynomials (e.g., Chebyshev, Jacobi) mitigate this with bound outputs and improved derivatives. Chebyshev-KAN approximates functions as follows:

$$\tilde{f}(x) = \sum_{j=1}^{d_{in}} \sum_{k=0}^n \Theta_{j,k} T_k(\tilde{x}_j) \quad (8)$$

where $\tilde{x}$ is the normalized input value of x , and $\tilde{x} \in [-1, 1]$, $T_k(\tilde{x}_j)$ represents Chebyshev polynomials and $\Theta \in \mathbb{R}^{d_{in} \times d_{out} \times (n+1)}$.

Instead of spline coefficients, the Naive-Fourier-KAN uses 1-D Fourier coefficients. By using the Fourier series, this layer approximates the KAN univariate functions. The definition of this layer is as follows:

$$f(x) = \sum_{k=1}^N a_k \cos(kx) + b_k \sin(kx) \quad (9)$$

Here a_k and b_k are Fourier coefficients, x is the input value. The number of Fourier terms (grid size) is N .

Recent advancements in enhanced KAN architectures focus on adaptive and hybrid designs. These designs integrate adaptive basis functions, attention mechanisms, and combinations of established deep learning components. The CKAN is one of the most basic structures. It combines the nonlinear modeling advantages of KAN with the spatial feature extraction capabilities of convolution operations. This breaks through traditional CNN performance bottlenecks in terms of feature representation and parameter efficiency, creating an overall optimization effect. Figure 2 shows the structure of a basic CKAN model.

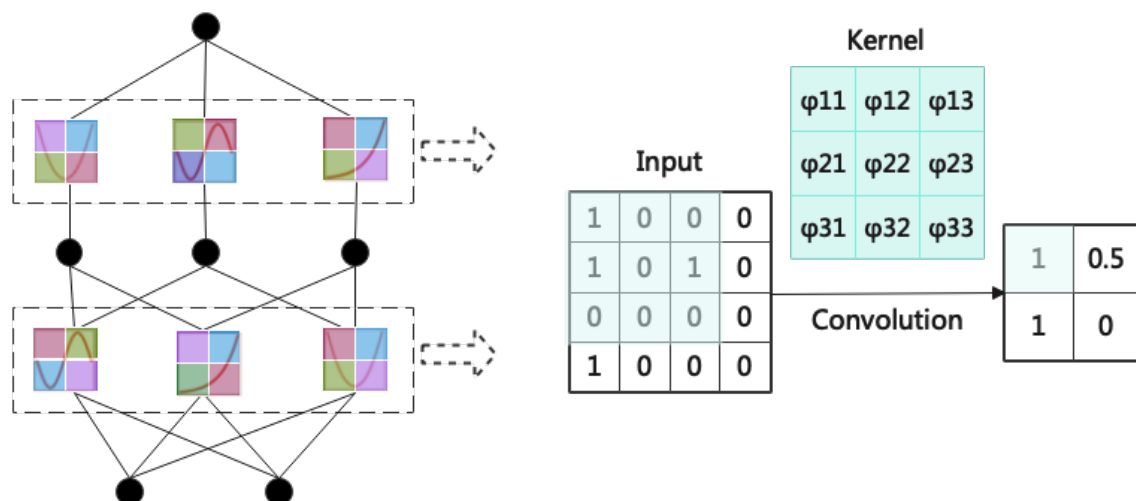


Figure 2. Basic CKAN model.

Another one, the KAN-MHA model, combines adaptive basis functions with MHA, enabling focused learning on critical flow regions (e.g., leading/trailing edges of airfoils) via parallel attention heads, significantly improving prediction accuracy and consistency in flow field modeling [26]. Figure 3 illustrates the overall structure of the model.

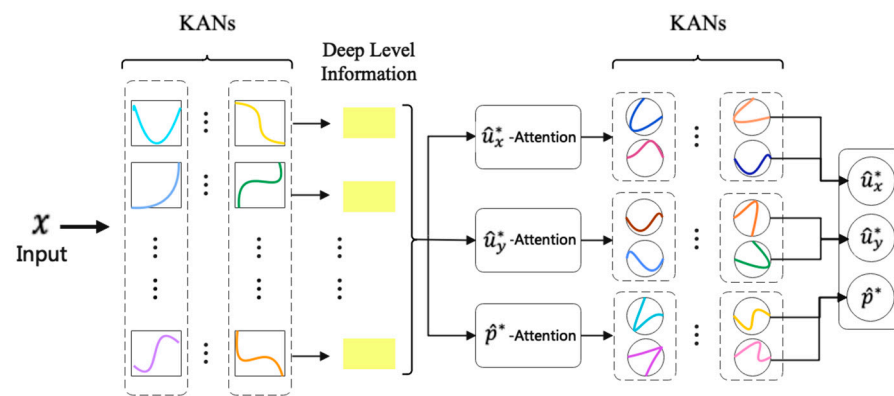


Figure 3. Multi-head attention KAN structure.

Hybridization with Transformers has emerged as a key strategy, with models like TFKAN and MTF-AViTK replacing traditional MLP layers in Transformers with KAN layers. For KAN layers' compact representation, TFKAN reduces parameter counts by 78% compared to conventional Transformers while maintaining high accuracy in IoT intrusion detection, demonstrating the efficiency of combining a Transformer with KAN's adaptive representations [9]. MTF-AViTK integrates KAN into an adaptive ViT for tool wear recognition, using self-attention for feature extraction and KAN in the classification layer to handle complex nonlinear mappings [27]. KANEFT further advances this by introducing Linear-KAN and Convolutional-KAN layers within a Transformer framework for hyperspectral image classification, where Convolutional-KAN enables multi-scale local feature extraction alongside spatial attention for global context integration [28].

For sequence and temporal data, hybrid KAN models with RNNs and TCNs have been developed. In BiLSTM-KAN, the temporal dependency capture of LSTM is integrated with the nonlinear feature extraction of KAN, thus increasing the robustness of time series forecasting, while TCN-KAN incorporates the multi-scale local dependency modeling of TCN with KAN's adaptive transformations to reduce overfitting when predicting electricity demand [29]. A KAN model combined with BiLSTM and TCN is shown in Figure 4.

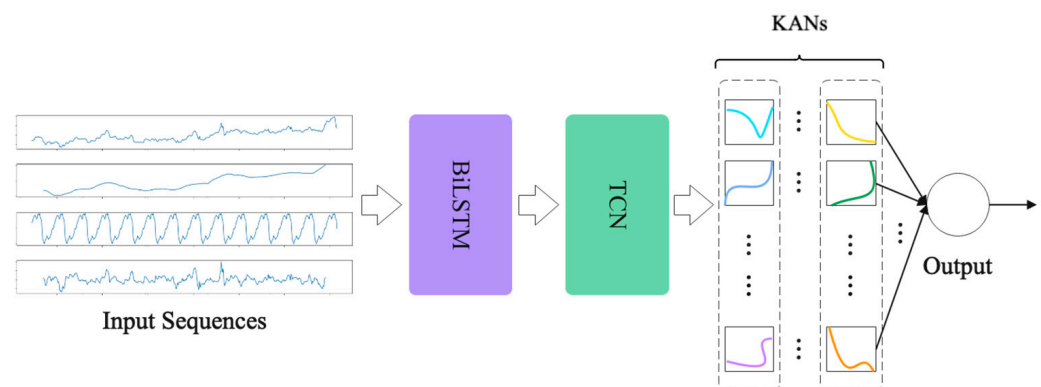


Figure 4. KAN model combined with BiLSTM and TCN.

Integration with GNNs is exemplified by MKAN-MMI, which incorporates KAN into a masked graph autoencoder for medicine–microbe interaction prediction. KAN enhances weight learnability and interpretability in the prediction layer. It outperforms current leading models in capturing complex nonlinear mappings within sparse biological datasets [30]. The GNN encoder obtains node embeddings based on the input graph, while the decoder reconstructs the graph based on a set of known relationships between nodes. Figure 5 illustrates the MKAN-MMI model's architecture.

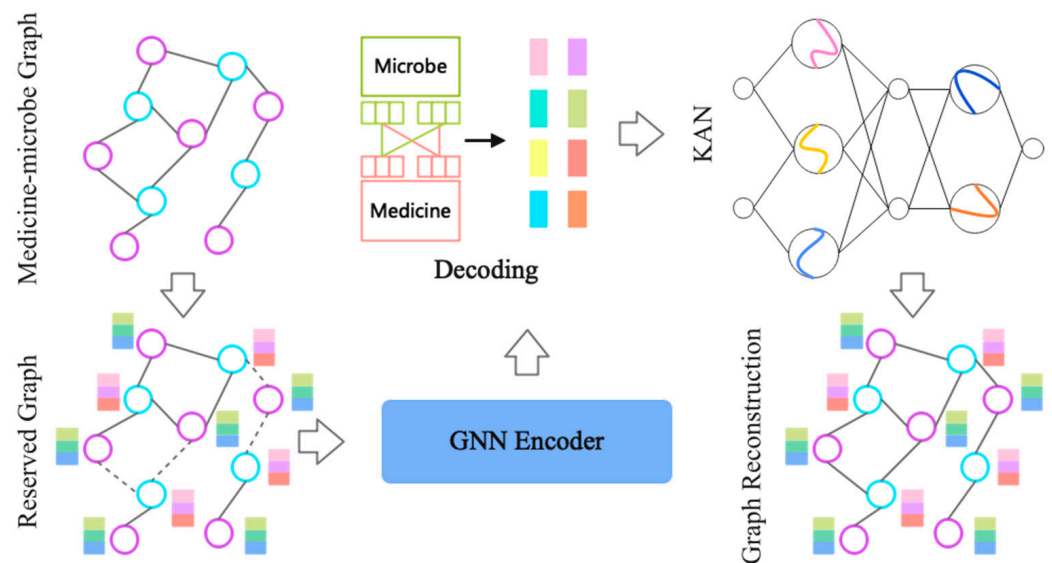


Figure 5. The MKAN-MMI model's architecture.

Ablation studies validate the efficacy of these designs: KAN-MSA in KAN- $\mathcal{T}$ improves PSNR compared to standard MSA [31]. As well as KAN layers in KANEFT [28] and HKAN [32] are critical for superior performance in hyperspectral image classification and fabric defect segmentation, respectively.

3. Applications of KAN in Scientific Computing and Engineering

3.1. Physics-Informed Neural Networks and Partial Differential Equations

In recent years, KANs have become increasingly integrated into physics-based machine learning frameworks for the solution of forward and inverse PDEs, which has led to the development of PIKANs, KINNs, and DeepOKANs [10]. By replacing traditional MLPs in PINNs and DeepONets, these models improve approximation capabilities for complex PDE solutions utilizing KANs' unique activation functions constructed from basis functions such as B-splines, orthogonal polynomials (e.g., Chebyshev, Legendre, Jacobi), radial basis functions, and wavelets [2].

An important advantage of KAN-based frameworks is their interpretability. The KANs use nested spline bases as activation functions, which provide a natural interpretation of the PDE solution space [2]. A KAN-based AI4S framework, FiberKAN, has been proposed to discover and characterize the underlying dynamics of fiber optic systems [8]. With its trainable, transparent, and interpretable activation functions, it can explain and understand complicated systems. For stochastic PDEs, Bayesian inference integration (e.g., Hamiltonian Monte Carlo in PIKANs, Bayesian Higher Order ReLU-KANs) can also provide epistemic uncertainty estimates [10,33].

KAN-based models are also computationally efficient, requiring fewer parameters than MLPs. Accordingly, PIKANs for power system DAEs can achieve high accuracy at a 50% reduction in network size compared to conventional PINNs [34]. There have been demonstrations of accelerated training using JAX-based implementations of PIKANs, which have achieved up to 84 times faster training than previously implemented KANs [6]. Figure 6 shows the model structure. However, due to unoptimized libraries and complex activation functions, some architectures require longer training time per epoch.

For fluid fields, KA-PointNet integrates KANs with PointNet in the irregular domains. The method predicts more accurately the pressure and velocity distribution along the surface of cylinders, resulting in more accurate calculations of lift and drag [35]. For turbulent flows, the AIVT method uses Chebyshev KANs to infer temperature and velocity fields

from sparse experimental data in Rayleigh-Bénard convection, achieving high resolution, offering an additional approach to studying thermal turbulence [36]. In topology optimization, the KATO method uses convolutional KANs to minimize stress, achieving 10% lower maximum stress and 67% reduced computational time in comparison to conventional approaches [37]. Hybrid models like AL-PKAN, combining GRU encoding with KAN decoding and augmented Lagrangian loss, achieve higher accuracy than traditional PINNs for PDEs [38]. The results of the analysis for these models are presented in Table 1.

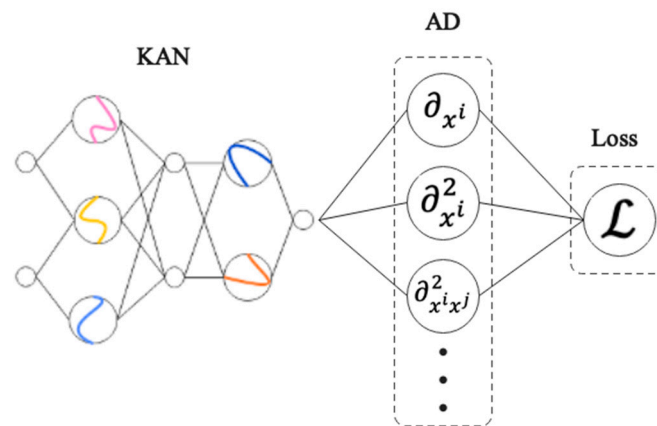


Figure 6. Schematic representation of a PIKAN.

Table 1. KAN-based model analysis for scientific computing and engineering.

Ref.	Model(s)	Metric(s)	Advantage	Can Be Improved
[2]	KINN	Relative Error, Displacement, Relative Error, Mise, Relative Error, Parameters and Training Time	Computes solid mechanics problems with significantly higher accuracy and convergence speed than MLP	Solving complex geometries
[6]	PIKANs	Parameters Number, Relative L^2 Error and Training Time	Enhance the solution accuracy, reducing L^2 error	Explore more appropriate optimizers and techniques for handling adaptive states in grid updates
[8]	FiberKAN	Parameters Number, Training Time, Inference Time, Absolute Error, and Operations Number	Improve performance by discovering and characterizing explicit, implicit, and non-analytical solutions	Low training and inference efficiency
[33]	ReLU-KANs	MSE, STD, and Training Time	Accurately identifying functional dependencies introduced by stochastic terms	High-dimensional, large-scale, and nonlinear dataset applications present challenges
[34]	PIKANs	Estimation Error	Enhance accuracy through a smaller network	Complexity of computing and ability to learn continuously
[37]	KATO	Stress Minimization	In addition to improving stress optimization, it also provides insight into topology design	Dimensionality restriction, interpretability, and vRAM consumption
[38]	AL-PKAN	Relative L^2 Error and MSE	Excellent generalizability and accuracy	Training time

3.2. Dynamical Systems Modeling and Scientific Discovery

KANs have demonstrated significant potential in dynamical systems modeling and scientific discovery, with applications spanning dynamical systems identification, neural

ODEs, and domain-specific scientific inquiry. In dynamical systems discovery, KANs address a critical limitation of traditional sparse optimization-based methods—their reliance on the often-unrealistic sparsity assumption that governing equations consist of only a few elementary functions [11]. As a general model-discovery framework, KANs decompose multivariate functions into sums of univariate functions, enabling accurate capture of complex nonlinear dynamics in systems lacking sparse structures. This capability extends to real-world scientific problems like modeling the AMOC fingerprint.

In the research of neural ODEs, KANs have been integrated into a specialized framework (KAN-ODEs) that replaces black-box MLPs with interpretable, parameter-efficient KAN structures [1]. Compared to traditional ODEs, KAN-ODEs offer higher accuracy, faster convergence, and superior interpretability. The framework works well in data-lean, nonlinear scenarios, successfully learning symbolic source terms and complete solution profiles for phenomena including wave propagation, shock formation, and the complex Schrödinger equation.

Within scientific discovery, KANs contribute to interpretable systems modeling by capturing relationships in domain-specific mechanisms. For example, the CardiOT model integrates KANs into a GNN framework for compound cardiotoxicity prediction, where KANs enhance both interpretability and accuracy [39]. This allows CardiOT to model relationships in cardiotoxicity mechanisms, with interpretability analyses (e.g., via EdgeSHAPer) linking KAN-learned weights to critical chemical effects, thereby advancing drug safety.

3.3. Load, Demand, and Environmental Forecasting

KAN-based models have demonstrated significant efficacy in power system load forecasting across diverse contexts. In district heating systems, the application of KAN enhanced heat load forecasting by dynamically adjusting spline function positions to capture local features and nonlinear relationships [40]. Based on real campus heating operational data, the model achieves a R^2 value of 0.98 and provides significant accuracy improvements over traditional methods at 3 h and 6 h timescales. For short-term load forecasting, a KAN model applied to a Swiss power system dataset outperformed MLPs and XGBoost by extracting periodic and random features [41]. Hybrid architectures integrating KAN with TCN, BiLSTM, and Transformer further improved electricity demand forecasting accuracy in the UK [29]. The TCN-KAN, Transformer-KAN, and BiLSTM-KAN models reduce RMSE and MAE while increasing R^2 compared to standalone LSTM/GRU models. In commercial buildings, KAN accurately predicted chiller energy consumption ($R^2 = 0.9465$, RMSE = 6.1023) by modeling nonlinear dynamics using cubic spline basis functions and advanced parameter identification methods, outperforming ANN and hybrid metaheuristic-deep learning models [42].

For DR prediction, the hybrid HITSKAN model combined KAN with N-HiTS to forecast day-ahead DR potential in residential buildings without requiring appliance-level data [43]. The HITSKAN architecture is shown in Figure 7. By leveraging KAN's ability to model complex nonlinear multivariate functions and N-HiTS's capacity to handle temporal dependencies via hierarchical interpolation and multi-rate sampling, HITSKAN outperformed state-of-the-art models (N-HiTS, TimesNet, Informer, TCN) across MAE, MAPE, RMSE, and sMAPE metrics in both the winter and summer seasons, with reduced computation time.

In water resource management, KAN has advanced hydrological forecasting and flood risk assessment. For river discharge forecasting, KAN models outperformed Transformers in short-term (1–3 days) predictions across the Rhine, Danube, Elbe, and Oder rivers [44]. In FSM, novel KAN variants (Boubaker-KAN, Cheby-KAN, VietaPell-KAN) achieved an

average overall accuracy of 92.5%, F1 score of 93.90%, and MCC of 0.866 in Iran's Karun and Gorganrud basins, outperforming Random Forest and XGBoost [45]. These models, by utilizing polynomial basis functions, effectively capture basin-specific nonlinear relationships between flood conditioning factors and flood occurrences, with fewer parameters (~4000–21,000) and faster processing times (~10 s).

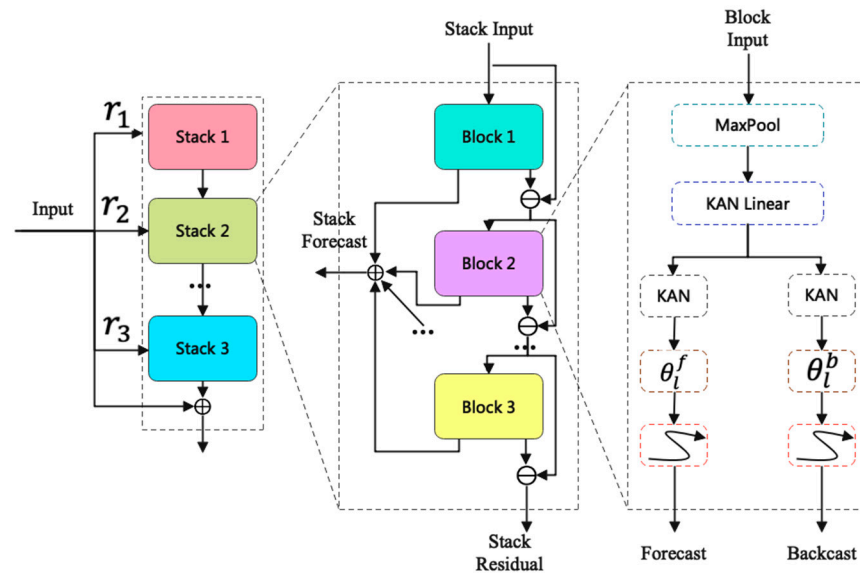


Figure 7. The HITSKAN architecture.

In environmental forecasting, KAN has enabled progress in cyanobacterial bloom monitoring, wind nowcasting, and weather impact assessment. For cyanobacterial monitoring in Canada's Pigeon Lake, KAN classified shoreline sections into blooming hot spots/non-hot spots using Dynamic World land cover data (500 m radius), achieving a recall of 0.83 [46]. For wind nowcasting at Madeira International Airport, KAN improved GFS outputs by applying nonlinear transformations to correct topographic effects, reducing MSE by 48.5% [47]. Additionally, KAN predicted airport flight operation resilience under weather conditions at Xi'an Xianyang International Airport, achieving $R^2 = 99.8\%$, $MAE = 0.0031$, and $MAPE = 2.3\%$ by modeling relationships between weather factors (temperature, wind speed) and operational metrics [48]. Table 2 presents the analysis results of these models.

Table 2. KAN-based model analysis for load, demand, and environmental forecasting.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[40]	Collected by the authors	R^2 , MAE, and RMSE	Outperforms conventional forecasting methods	Consideration of external factors
[41]	[49,50]	RMSE, MAE, and MAPE	With identical parameters, it surpasses MLP	STLF requires more stable methods for generating symbolic equations
[29]	Collected by the authors	R^2 , MAE, and RMSE	Enhances the performance of multi-step forecasting	Balancing prediction quality and computational effort
[42]	[51]	R^2 , MAE, MAX, and RMSE	Superior accuracy and practical utility	Quality of data, the interpretability of the model, and system integration

Table 2. Cont.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[43]	[52]	MAE, MAPE, RMSE, and sMAPE	Outperforms state-of-the-art models	Adaptability to diverse and complex energy consumption
[44]	Compiled by the authors	KGE, R^2 , MAPE, and RMSE	Superior short-term prediction capacity	Insufficient long-term forecasting ability and generalizability
[45]	Compiled by the authors	Accuracy, Precision, Recall, F1 score, Cohen's Kappa, Balanced Accuracy, MCC, and Jaccard Score	Superior performance in key metrics	Generalizability and practicality need improvement
[46]	Dynamic World dataset	RMSE Train, RMSE Test, Recall, Precision and R^2	Optimizes computational budgets significantly	Model mechanisms need to be further explored
[47]	GFS and LPMA-MID	MAE, MSE, RMSE, and R^2	Transparency, explainability, and interpretability are improved	Enhancing the efficiency of computation
[48]	Compiled by the authors	MSE, RMSE, MAE, MAPE and R^2	Exceptional predictive ability	Further validation in extreme weather conditions

4. Applications of KAN in Medical and Biological Domains

4.1. Medical Image Segmentation and Classification

KAN-integrated frameworks have been extensively developed for various medical imaging tasks, addressing challenges such as nonlinear feature modeling, boundary delineation, and noise robustness. In MRI cardiac segmentation, Li & Xu [53] proposed KS-Net, a semi-supervised U-shaped architecture incorporating KAN modules and a dual-stream perturbation strategy. Based on MyoPS 2020 and ACDC datasets, KS-Net is capable of segmenting the left ventricle, myocardium, and right ventricle with high Dice scores. Ablation studies confirm the KAN module's critical role in enhancing nonlinear knowledge sharing and segmentation accuracy.

For lung nodule detection in CT scans, Jiang et al. [54] developed KansNet, which integrates a PKAB into a 3D CNN encoder–decoder network. Equipped with MSPCA and AMSF modules, KansNet outperformed state-of-the-art methods on the used dataset. Achieved a CPM of 90.32% and an AP of 89.19% on the LUNA16 dataset, with notable improvements in detecting small nodules (3–10 mm).

In medical image segmentation, two prominent frameworks have been introduced: KAC-Unet and MM-UKAN++. Lin et al. [55] proposed KAC-Unet, which combines KAN with an AGS and COT attention mechanism within a Unet architecture. It achieved Dice scores of 78.63%, 89.25%, and 92.55% on breast ultrasound (BUSI), dermoscopy (ISIC), and prostate MRI datasets, respectively. Zhang et al. [56] presented MM-UKAN++, replacing convolutional blocks with multilevel KAN layers and incorporating a FCSA mechanism. This framework delivered 69.42% IoU and 81.30% Dice, with real-time performance (40.8 FPS) and superior boundary delineation for regions of interest.

For computing tomography analysis, Chau et al. [57] applied KANs to the binary classification of QCT data for respiratory conditions induced by occupational cement dust exposure. Using 141 QCT-derived variables, their two-hidden-layer KAN model achieved 98.03% accuracy, outperforming traditional machine learning and deep learning models, with SHAP analysis identifying key features. Additionally, Amin et al. [58] introduced

ViT-KAN, which integrates KAN layers into a Vision Transformer for lung EIT image reconstruction using CT scans as ground truth. This framework achieved a mean squared error of 0.0045 and a structural similarity index of 0.96, demonstrating robustness to Gaussian noise and potential for clinical lung function monitoring. Table 3 summarizes the key information from the literature.

Table 3. Applications for medical image segmentation and classification.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[53]	ACDC and MyoPS 2020	Dice Score, Jaccard Score, Hausdorff Distance, and FLOPs	Outperforms existing state-of-the-art approaches	Hyperparameter selection requires further explanation
[54]	LUNA16 and TIANCHI	Sensitivity, False Positives, AUC, AP, and CPM Scores	A highly promising auxiliary diagnostic tool	No relevant analysis of noise data
[55]	[59–61]	IoU, Dice, Recall, Accuracy, and Precision	Outperforms state-of-the-art models	Lack of discussion on hyperparameters
[56]	BUSI, DDTI, CUBS, and CAMUS	Dice Score, IoU, HD95, FLOPs and Parameter Quantity	Low-cost segmentation with high accuracy	Convergence speed can be improved
[57]	Collected by the authors	Accuracy, Precision, Recall, F1 score, Specificity, and MCC	All metrics show high performance	Adding external validation to its decision would be beneficial

4.2. Physiological Signal Processing and Health Monitoring

In EEG seizure detection, the KAN-based model has demonstrated efficacy in balancing accuracy and parameter efficiency. KAN-EEG, for example, employs spline-parametrized learnable activation functions to achieve high out-of-sample generalization across diverse datasets. With high AUROC values while requiring fewer network parameters and less training data compared to other models [62]. Another study comparing KAN with LSTM for EEG seizure prediction, however, found that the LSTM model performed better than KAN [63].

For PPG signal anomaly recognition, an architecture integrated parameterized learnable activation functions has been shown to strike a favorable balance between accuracy and parameter efficiency. Compared to the high-performing but parameter-heavy CNN-LSTM model (95% accuracy, 456,865 parameters), the CNN-KAN model achieved competitive performance (89% accuracy, AUC 95%) with substantially fewer parameters (127,206) [64].

In blood glucose prediction, the SugarNet model, which integrates frequency-domain learning with a Fourier Kolmogorov–Arnold network, excels at modeling individualized glucose dynamics. In comparison with state-of-the-art approaches, it demonstrated superior performance. Its balance of accuracy and parameter efficiency makes it promising for integration into diabetes management systems, including proactive glycemic control and autonomous insulin delivery [65].

Beyond these specific tasks, KANs have shown advantages in broader physiological signal-based health monitoring applications. For instance, in multi-task sitting posture recognition, a KAN model achieved higher average accuracy (97.03% upper body, 92.11% lower body) with fewer parameters (4320) than an MLP (93.87% upper body, 91.47% lower body; 7662 parameters) [66]. Providing evidence of KANs' ability to provide accurate and efficient health monitoring. Additionally, methodological advances in KAN-based frameworks, such as PKAN for peptide function and activity prediction [67]. By eschewing linear weights and using learnable activation functions, the method is improved to handle

multimodal peptide language information nuancedly. Table 4 summarizes and analyzes the models mentioned in this section.

Table 4. KAN-based models in physiological signal processing and health monitoring.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[62]	TUH, EPILEPSIAE and RPAH	AUROC, AUPRC, Precision, Recall, and F1 score	Resilience, versatility, and high-level generalization	Further exploration of more optimal architectural designs
[64]	Collected by the authors	Accuracy, Precision, Recall, F1 score, and AUC	Reduces the number of parameters while improving performance	Efficiency metrics were not considered
[65]	[68–70]	RMSE and MARD	Outperforms all other methods consistently	Glycemic abnormalities are not sufficiently detected
[66]	Collected by the authors	Accuracy	Higher accuracy	Inference speed is slower than MLPs
[67]	[71,72]	SE, SP, ACC, MCC, AUC, MSE, RMSE, MAE, and R^2	Outperforms state-of-the-art models with a streamlined design	Chemical structural complexity isn't fully represented

5. KANs in Cybersecurity and Industrial Applications

5.1. Intrusion Detection Systems in IoT and Network Security

KANs and hybrid KAN models have demonstrated significant potential in IDS for IoT and network security, effectively addressing cyberattack detection, botnet classification, and anomaly detection. CKAN architectures, which integrate KAN layers into CNN frameworks (such as [73]) by replacing traditional MLP layers, have shown superior performance across multiple benchmark datasets. For example, one CKAN model outperformed conventional deep learning models like CNN, RNN, and Autoencoder in both accuracy and stability [13]. Similarly, another CKAN variant with attention mechanisms, designed for IoT and IoMT cybersecurity, exhibited enhanced accuracy on CICIoT2023 and CICIoMT2024 datasets, though it faced limitations in memory usage and inference speed (88 samples/sec) due to spline-based kernel computations [74]. SNN-based intrusion detection is significantly more efficient [75]. Transformer-based KAN model, which replaced MLP layers in Transformers with KAN layers, further extended its performance, achieving high accuracy scores of 99.96% (RT-IoT2022), 98.43% (IoT23), and 99.27% (CICIoT2023) while maintaining effectiveness in detecting diverse attacks [9]. Figure 8 shows the model flowchart.

In botnet classification, various KAN architectural variations (Original-KAN, Fast-KAN, Jacobi-KAN, Deep-KAN, Chebyshev-KAN) outperform traditional ML and DL models (e.g., MLP, CNN, LSTM, GRU) across datasets including N-BaIoT, IoT23, and IoT-BotNet. In particular, the Original-KAN is exceptional at capturing complex, nonlinear features present in IoT botnet traffic, resulting in higher accuracy and faster convergence [76]. For automotive cybersecurity, the DSC-KAN model, combined with SN-GAN for synthetic anomaly data generation, demonstrated robust detection of CAN bus attacks, outperforming CNNs and Vision Transformers while reducing computational cost [77]. Table 5 summarizes the key information from these studies.

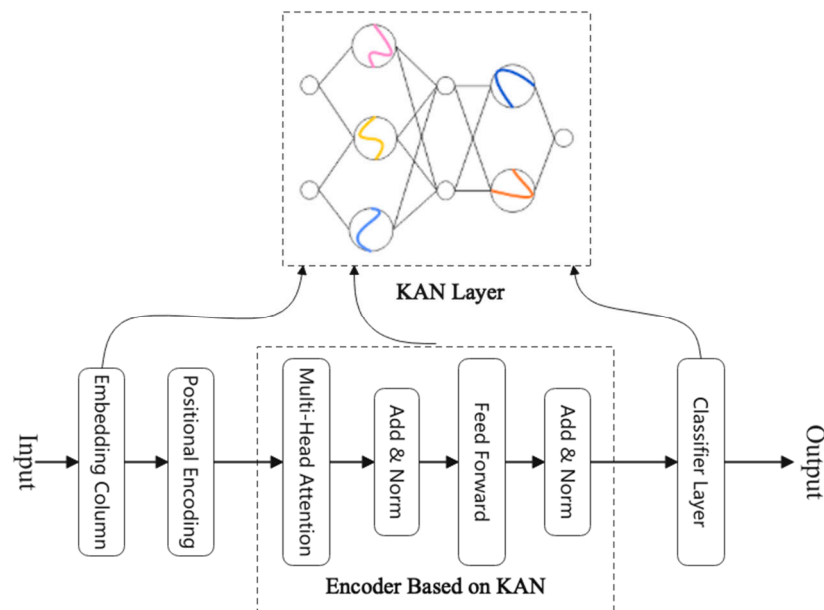


Figure 8. TFKAN model flowchart.

Table 5. KAN-based models for intrusion detection systems in IoT and network security.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[13]	NSL_KDD, CICIoT2023 and TONIoT	Accuracy, Precision, Recall, and F1 score	Performs better than other DL models	Reduce the processing time required for training
[74]	CICIoT2023 and CICIoMT2024	Accuracy, Precision, Recall, F1 score, False Positive Rate, Execution Speed, Energy Consumption, RAM, Model Size, Parameters, and FLOPs	Detection accuracy metrics are better	Model efficiency needs to be improved
[9]	RT-IoT2022, IoT23 and CICIoT2023	Accuracy, Precision, Recall, and F1 score	Detection of intrusions with high accuracy	Reduce the processing time required for training
[76]	N-BaIoT, IoT23 and IoT-BotNet	Accuracy, Training Time, Precision, Recall, and F1 score	Outperform consistently	Training requires significant computational power
[77]	[78]	Precision, Recall, and F1 score	The potential for significant improvement in vehicle network security	Efficiency metrics were not considered

5.2. Industrial Fault Diagnosis and Structural Monitoring

KANs have demonstrated significant potential in industrial fault diagnosis and structural monitoring, with applications spanning machinery components, power transformers, and power electronic systems, characterized by enhanced interpretability and suitability. In machinery component fault detection, KAN-based frameworks have been effectively applied to tool wear recognition and bearing fault diagnosis. A ViT that incorporates KAN (MTF-AViTK) can recognize tool wear states in manufacturing by converting 1D time series cutting signals into 2D MTF images in order to identify long-term dependencies. This data is then extracted using the Adapt-ViT module, and it is classified using a two-layer KAN. Validated on the PHM2010 dataset, MTF-AViTK achieved high classification accuracy with millisecond-level inference time, supporting real-time deployment in industrial

settings [27]. Similarly, for bearing fault diagnosis, a KAN-based framework automates feature selection and hyperparameter optimization using shallow architectures, enabling lightweight deployment on resource-constrained edge devices [79]. This framework yielded perfect or near-perfect F1-Scores on the CWRU and MaFaulDa datasets.

In power transformer fault diagnosis, KANs have been tailored to address imbalanced real-world data. The KAN_{Diag} approach employing the SyMProD oversampling technique to balance datasets and KAN layers with trainable spline function matrices for classification [80]. Compared to other models, KAN_{Diag} achieved superior performance and improved interpretability.

For power electronic components, such as VSCs in power systems, DropKAN has emerged as a robust data-driven impedance modeling approach. DropKANs outperform conventional FNNs, LSTMs, and standard KANs in impedance identification accuracy, training/prediction times, and model simplicity [81]. Table 6 analyzes and summarizes the models mentioned in this section.

Table 6. KAN-based models for industrial fault diagnosis and structural monitoring.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[27]	PHM2010	Accuracy, Precision, Recall, Specificity, and F1 score	Enhanced robustness and generalizability	Consider multi-dimensional signals
[79]	CWRU and MaFaulDa	F1 score	Exceptional performance in identifying faulty conditions	The evaluation metric is limited
[80]	TAESA	Hamming Loss, F1 score, and Accuracy	Robustness and efficiency	Consider the different types of transformer damage
[81]	Simulated dataset	Training Time, Prediction Delay, and MSE	Simplicity, accuracy, and training and prediction times	Stability and robustness require further analysis

6. KANs in Image Processing and Vision Tasks

6.1. Hyperspectral Image Classification and Fusion

Through innovative architectural designs and superior nonlinear feature modeling capabilities, KANs have demonstrated substantial contributions to functions such as hyperspectral image classification, fusion, change detection, and spatiotemporal remote sensing. Jamali et al. developed a HybridKAN architecture combining 1D, 2D, and 3D KAN models, which achieved competitive classification accuracy compared to state-of-the-art CNN and Vision Transformer models, with faster convergence and improved feature separability visualized via t-SNE [82]. Similarly, Han et al. proposed KANEFT, a KAN-driven Enhanced Fusion Transformer that leverages KAN layers for linear/convolutional operations and a Multi-scale Fusion Learning Transformer module, achieving overall accuracies up to 99% on datasets like Indian Pines and Pavia University [28]. Firsov et al. introduced HyperKAN, replacing traditional layers with KAN blocks in architectures such as 1D-CNN, 3D-CNN, and transformers, leading to consistent accuracy improvements across seven hyperspectral datasets, enabling better modeling of complex patterns [7].

In hyperspectral image fusion, Zhu et al. proposed SFIGNet, a KAN-based spatial-frequency dual-domain implicit guided sampling network for HMIF. By integrating KAN with INR, SFIGNet conducts guided implicit sampling in spatial and frequency domains via SIGSU and FIGSU units, achieving superior spatial-spectral fidelity with a lightweight design [83].

In spatiotemporal remote sensing applications, Wang et al. proposed UKAN, a KAN-integrated U-Net, for flood inundation extraction from multisource data. UKAN outperformed U-Net and TransUNet on flood datasets, with KAN layers enhancing sensitivity to noise and low-contrast regions [84].

For change detection, a critical task in remote sensing, Liu et al. developed AEKAN, a superpixel-based AutoEncoder KAN for unsupervised multimodal change detection in HRSIs [85]. Using a Siamese KAN encoder with dual KAN decoders and hierarchical commonality loss, AEKAN outperformed state-of-the-art methods on five MCD datasets validated via ROC, PR curve, and metrics like overall accuracy and Kappa coefficient. In a study of HCD, Seydi et al. found that parameter numbers are less important in KAN-based models than nonlinear functions and model structure [12]. The Chebyshev-KAN model achieved an average overall accuracy of 97.35% across datasets, surpassing deep learning models like the Swin Transformer. Table 7 analyzes the models mentioned in this section.

Table 7. KAN-based model for hyperspectral image classification and fusion.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[82]	QUH-Pingan, QUH-Tangdaowan, and QUH-Qingyun	Overall accuracy, Average accuracy, Kappa accuracy, Per-class accuracies	Outstanding overall performance	Some datasets have inferior accuracy
[28]	Indian Pines, Pavia University, and Houston University	Model complexity, Overall accuracy, Average accuracy, Kappa accuracy, Per-class accuracies	Overall accuracy is remarkable	Complexity of the model needs to be improved
[7]	PaviaU, PaviaC, Salinas, Indian Pines, Houston 13, Houston 18, KSC	Overall accuracy, Average accuracy, Weighted F1 measure	Overwhelmed traditional neural networks consistently	Efficiency metrics were not considered
[83]	PaviaU, DFC2018, Botswana and Houston	RMSE, PSNR, ERGAS, SAM, Model complexity	Outperforms comparative algorithms on a variety of scale factors	Efficiency of training and consideration of extreme noise conditions
[84]	C2S-MS Floods and MMFlood	IoU, Precision, Recall, F1	Outstanding performance with high parameter and computational efficiency	Fails to recognize flood areas, particularly simple ones
[85]	Five MCD datasets	ROC, PR, AUC, AUP, OA, Ka, F1	Superiority on the MCD task and the ability to identify changes between HRSIs	Further testing is needed to verify stability, large scales, and complex scenes
[12]	Jiangsu Farmland Area, Jiangsu Inundated Area, Oregon Agricultural Area, and Kansas Agricultural Area	Accuracy, Precision, Recall, F1 score, Cohen's Kappa, Balanced Accuracy, Matthews Correlation Coefficient, and Jaccard Index	Outperforms deep learning methods while having a markedly lower complexity	Analyzing potential enhancements in model optimization or evaluating hybrid architectures

6.2. Low-Light Image Enhancement and Deblurring

For low-light image enhancement, the KSID method introduces KANs into low-level vision tasks to model nonlinear dependencies between low-light and normal images [86]. Embedding a KAN-Block—composed of KAN-Layers with spline-based convolutional layers and learnable activation functions—into a U-Net denoising network within a diffusion model framework. KAN-Blocks are demonstrated to play a significant role in learning nonlinear degradation relationships through ablation studies. In image deblurring, the

KT-Deblur framework integrates KANs with transformer architectures by replacing fixed activation functions with dynamically learnable ones based on KANs, enabling adaptation to various blur types [87]. Combining a U-Net architecture with a GAN framework and a UAFE module, it optimizes noise suppression and detail enhancement by combining neighborhood self-attention with linear self-attention. Experimental results on datasets demonstrate outperformance over state-of-the-art methods. According to the authors' experimental results, the proposed KAN-based tracking method also demonstrates excellent effectiveness and robustness [88]. Figure 9 shows the model structure. Table 8 analyzes the models mentioned in this section.

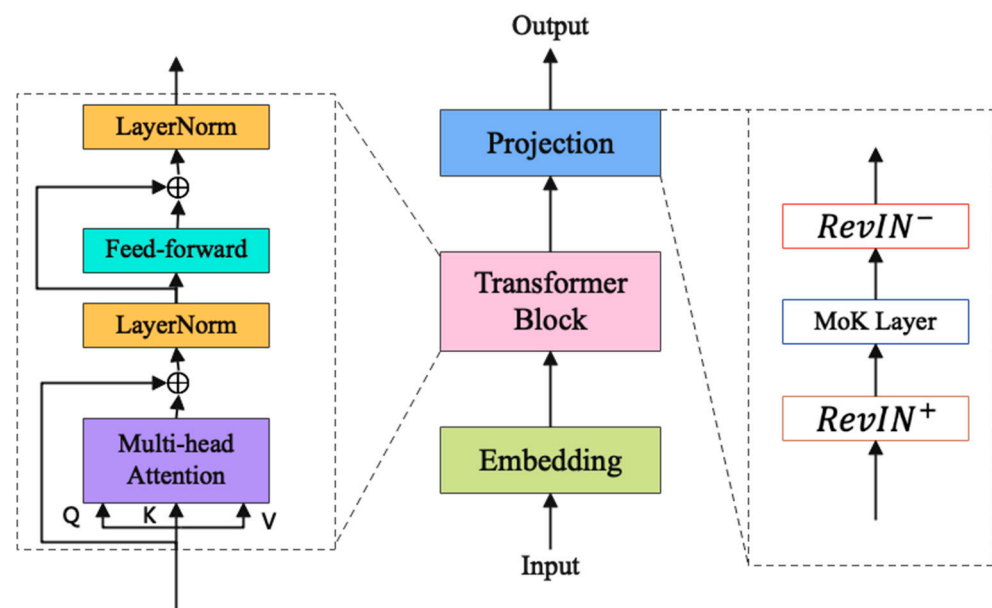


Figure 9. Structure of the KAN-based tracking model.

Table 8. KAN-based models for low-light image enhancement and deblurring.

Ref.	Dataset(s)	Metric(s)	Advantage	Can Be Improved
[86]	LOLv1, LOLv2 and LSRW	PSNR, SSIM, LPIPS, FID, Model Parameter Count, and Average Inference Time	The method performs well on benchmark datasets	Robustness needs improvement
[87]	RSML_C, RSML_S, GoPro and RealBlur	PSNR and SSIM	Performance superior to others	Efficiency metrics were not considered
[88]	Collected by the authors	MAE, MSE, MOTP, and MOTA	State-of-the-art performance	Lighting conditions and moving objects can affect performance

7. Challenges, Limitations, and Future Research Directions

KANs face a wide range of challenges in a number of critical areas. Robustness issues are evident, such as insufficient robustness in tool wear conditions recognition [27], susceptibility to adversarial attacks and noise with a risk of underperformance compared to MLPs in complex or noisy conditions [89], and persistent noise resilience concerns despite partial robustness demonstrations [11]. Ultrasound image segmentation suffers when regions of interest have low contrast with the surrounding tissues [56].

KAN replaces MLPs' linear weights with learnable nonlinear basis functions. This offers significant theoretical advantages in terms of function approximation accuracy and interpretability. However, realizing these benefits often relies on optimizing the basis

function parameters, which increases computational costs. As a result, KAN remains unsuitable for scenarios where the data itself demands substantial computational power. During application, additional considerations such as hardware configuration and energy consumption may be required.

Optimization instability poses another major hurdle. The loss of PIKANs increases sharply after grid extensions due to the re-initialization of the optimizer state [6]. When the KAN module is removed from models like KRN-DTI, performance may be significantly reduced [90]. PINNs are also prone to instability due to penalty factor inflation [38].

Scalability challenges remain; larger KANs may impede interpretability, KAN technology exhibits high computational complexity, and KAN-based topology optimization models, such as KATO, require large amounts of vRAM due to dense tensor representations, restricting high-resolution applications [37]. While variants like KAN-Mixer and convolutional KANs have been explored, the scalability results remain mixed.

Handling complex data geometries is problematic, with KANs struggling under adversarial attacks on complex geometries. US images present challenges like low contrast, fuzzy boundaries, and scale variations. Additionally, tuning hyperparameters, including grid sizes and spline orders, is critical, with neural network hyperparameters affecting optimization results significantly.

For future research, the summary of the aforementioned studies indicates that KAN's training efficiency is low. This necessitates the optimization of basis functions and algorithms to improve computational efficiency. Research into lightweight KAN variants suitable for edge devices holds significant practical value, as it advances their real-time application to intelligent sensors, edge controllers, and similar equipment. Moreover, the current selection and combination of basis functions in KAN remain relatively fixed. Future research could explore dynamic adaptive mechanisms for basis functions to enhance multi-scale, complex nonlinear feature capture. Further exploration of fusion innovation between different model architectures may be pursued, leveraging their respective strengths to construct high-performance frameworks tailored to specific domains.

8. Conclusions

The KAN framework is underpinned by the Kolmogorov–Arnold representation theorem as its core theoretical foundation, whereas MLP is constructed upon the universal approximation theorem. Based on the Kolmogorov–Arnold Representation Theorem, which states that any multivariate continuous function can be decomposed into a finite superposition of univariate functions, the core of the KAN architecture lies in the combined learning of univariate functions. Based on the Universal Approximation Theorem, a feedforward network with a single hidden layer and a nonlinear activation function can approximate any continuous function on a compact set with arbitrary precision. This provides the theoretical basis for MLP's ability to approximate functions through fixed nonlinear activation combined with weight adjustment. Among diverse function classes, including smooth functions and those derived from physical equations, KAN demonstrates distinct advantages owing to its unique structural design. With this structural innovation, nonlinear transformation and weight parametrization are integrated into unified layers. This leads to improved functional representation capability, flexibility, and interpretability. It has been shown that KANs extended with adaptive basis functions, hybridization with Transformers, CNNs, RNNs, and graph structures improve accuracy and parameter efficiency in a variety of domains. These domains include signal processing, physical modeling, and cybersecurity. Ablation analysis shows that LKAN can improve deep learning models with a reduction in parameters [91].

The use of KANs to diagnose and monitor industrial faults exploits their interpretability and efficiency for real-time condition assessments of machinery, transformers, and power electronics. There are also numerous image processing applications, such as medical image analysis, low-light image enhancement, etc. There are, of course, applications in other fields, such as stock price forecasting [92] and dam deformation prediction [93]. For quantum architecture search, KANs demonstrated $2\times$ to $5\times$ higher success probability than MLPs in preparing Bell and GHZ states under noiseless conditions and better fidelity in noisy environments [94]. In fraud detection, KANs' hierarchical structure provides a more effective and interpretable solution over MLPs, with Efficient KAN achieving a 100% F1 score and 0.1 s detection time [95]. In [96], CoxKAN is described as a Cox proportional hazards KAN designed for high-performance, interpretable survival analysis. Based on real datasets, CoxKAN consistently outperformed the traditional Cox proportional hazards model. Multi-exit KAN is another interesting model [97]. By including a prediction branch at every layer, the network can make accurate forecasts at multiple depths at the same time. It provides scientists with a practical method for high performance and interpretability. As the aforementioned examples illustrate, KAN can be used across a wide range of fields and generate valuable results.

According to comparative evaluations, KANs have many aspects of outstanding performance compared to mainstream neural architectures in terms of accuracy, parameter efficiency, interpretability, and training dynamics, although there is a trade-off in computational complexity and occasional optimization instability. It should be noted that some studies are based on datasets collected by the authors themselves. This may pose obstacles for researchers who need to reproduce the results. In summary, KANs are theoretically based and practically versatile neural networks with demonstrated applications in scientific computing, engineering, medicine, environmental, and cybersecurity. For KANs to fully realize their potential in complex function approximation and interpretable machine learning, continuous innovations and systematic investigation into their limitations are essential.

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Abbreviations

The following abbreviations are used in this manuscript:

KAN	Kolmogorov–Arnold network
MLP	Multi-Layer Perceptron
MHA	Multi-Head Attention
ViT	Vision Transformer
GNN	Graph Neural Network
MMI	Medicine–Microbe Interaction
GAE	Graph Auto Encoder
PDE	Partial Differential Equation

DAE	Differential-Algebraic Equation
AIVT	Artificial Intelligence Velocimetry–Thermometry
ODE	Ordinary Differential Equation
AMOC	Atlantic Meridional Overturning Circulation
ANN	Artificial Neural Network
DR	Demand Response
SE	Sensitivity
SP	Specificity
ACC	Accuracy
MSE	Mean Squared Error
RMSE	Root Mean Squared Error
MAE	Mean Absolute Error
AUC	Area Under Curve
AUP	Areas Under PR Curve
MOTP	Multiple Object Tracking Precision
MOTA	Multiple Object Tracking Accuracy
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity
LPIPS	Learned Perceptual Image Block Similarity
FID	Fréchet Inception Distance
OA	Overall Accuracy
PR	Precision–Recall
ROC	Receiver Operating Characteristic
Ka	Kappa coefficient
MAX	Maximum error
KGE	vKling-Gupta Efficiency
MARD	Mean Absolute Relative Difference
AUROC	Area Under the Receiver Operating Curve
IoU	Intersection over Union
FSM	Flood Susceptibility Mapping
MCC	Matthews Correlation Coefficient
GFS	Global Forecast System
PKAB	Partial Kan Attention Block
MSPCA	Multi-Slice Partition Channel Priority Attention
AMSF	Adaptive Multi-Scale Feature Fusion
CPM	Competition Performance Metric
AP	Average Precision
AGS	Adaptive Group Strategy
FCSA	Frequency-Channel Spatial Attention
EIT	Electrical Impedance Tomography
IDS	Intrusion Detection System
ML	Machine Learning
DL	Deep Learning
US	Ultrasound
CT	Computed Tomography
MTF	Markov Transition Field
VSC	Voltage-Source Converter
FNN	Feedforward Neural Network
LSTM	Long Short-Term Memory network
HMIF	Hyperspectral and Multispectral Image Fusion
INR	Implicit Neural Representation
HRSI	Heterogeneous Remote Sensing Image
HCD	Hyperspectral Change Detection
UAFE	Union Attention Feature Extraction

HSI	HyperSpectral Image
FCN-KAN	Fully Connected Network KAN
RBF-KAN	Radial Basis Function KAN
CKAN	Convolutional Kolmogorov–Arnold Network
PIKAN	Physics-Informed KAN
KINN	Kolmogorov–Arnold-Informed Neural Network
DeepOKAN	Deep Operator KAN
PINN	Physics-Informed Neural Network
DeepONet	Deep Operator Network
DSC-KAN	Depthwise Separable Convolutional KAN
SN-GAN	Spectral Normalization GAN
SAR	Success Attack Rate
STD	Standard Deviation
FLOP	Floating-Point Operation
LKAN	Linear KAN
HKAN	Hybrid Kolmogorov–Arnold Network
MSA	Multi-headed Self-Attention
KANEFT	Kolmogorov–Arnold Network-based Enhanced Fusion Transformer

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